# Accumulation–Release Asymmetry

## A Geometric Relational Framework

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# Revision statement

The April 2026 preprint introduced Accumulation–Release Asymmetry as a temporal phase ratio and proposed the golden ratio as the attractor of free-running self-organising systems. The framework has since been generalized into a whole-first relational geometry with explicit scale, orientation, child-to-parent projection and multi-axis decompression. Its principal physical proposal is that this recursively coupled wave/sphere relation is a bedrock geometry that generates larger identities and their observable relations. That proposal is larger than present testing capabilities and remains open. Frozen follow-up tests also did not support standalone Phi as a universal attractor or fixed radial endpoint. This version preserves the historical record while separating the stable coordinate framework, crosswalks to established science, empirical results and open physical hypotheses.

# Abstract

Accumulation–Release Asymmetry (ARA) is a research framework for representing a declared coupled identity as a wave or sphere and minimally measuring it through an oriented diameter from `0` to `2`. The coordinate is relational: its meaning depends on a published boundary, orientation, rung, time window and set of retained relations. The midpoint `1.0` is the ridge or core of the selected parent cut, while the endpoint labels reverse with orientation. **TE-ARA is not another geometry:** it is the bookkeeping name for the recurring normalized total-allocation view of that same ARA identity, whose native total is `2`, with physical magnitude retained separately. A complete child remains total-`2` on its own rung but its selected expression contributes half-weight when read one pure octave upward. Information$^3$ supplies the framework's minimum relational closure; two persistent perpendicular ARA relations form Di-ARA, and their ternary relations can be decompressed into an ARA$^9$ neighbourhood. The principal physical hypothesis is that this recursively coupled geometry is generative bedrock: larger identities and observable laws arise from repeated coupling, compression and decompression of the same relation. Directly proving an ultimate bedrock is beyond present measurement. The paper therefore tests the proposal indirectly by crosswalking established science, attempting predeclared structural recoveries and making frozen empirical predictions. A crosswalk demonstrates compatibility, not proof of generation. This paper records both supportive and failed applications across mechanics, field theory, quantum representations, prime structure, complex time series and learned systems. The universal physical hypothesis remains open. A two-axis irrationality programme based on contraction/expansion and forward/reverse traversal is retained as a testable extension; no universal Phi, exponential or fitted radial endpoint has yet been established.

# 1. Central proposal, evidential boundary and relation to the April 2026 preprint

## 1.1 The bedrock generative-geometry hypothesis

ARA proposes that the recurring two-pole wave/sphere relation is not merely a descriptive coordinate imposed on systems after they form. The larger hypothesis is that it is **generative and fractal**: the same relational geometry recurs within identities, between coupled identities and across scale, where repeated coupling forms children, parents, cross-rung relations and the observable structures later described by domain-specific physical laws.

Here *fractal* means recursive structural self-similarity: a completed ARA identity can contain smaller ARA relations and can itself participate as one component of a larger ARA relation. It does not assert that every physical realization has an exact statistical fractal dimension or one universal spatial appearance; those are separate empirical questions.

The wave and sphere descriptions are two views of that same recursion. At one selected slice, an oriented diameter records the Phase A/Phase B mixture as a position on `0-2`. As the sphere advances, rotates or is extruded through time, successive diameter readings trace a wave around the `1.0` ridge. Completion of that two-sided wave permits the whole identity to participate in a larger wave, while decompression reveals smaller waves inside it. The hypothesis is therefore not that circles, waves and rungs are three independent mechanisms: a wave is the time-unfurled trace of the sphere, and a rung is that completed wave read as a child of a larger ARA identity.

This is the framework's principal physical claim and its largest evidential burden. The definitions in this paper show that the proposed geometry can be stated consistently. They do not prove that nature universally instantiates it.

## 1.2 The recovery programme

The hypothesis is approached indirectly through an evidence ladder:

1. **Exact crosswalk:** represent an established equation in declared ARA coordinates without changing its predictions.
2. **Structural recovery:** use predeclared ARA relations to recover independently known architecture without inserting the answer after inspection.
3. **Empirical discrimination:** make a frozen prediction on untouched data that is not guaranteed by normalization and can lose to named controls.
4. **Cross-domain recurrence:** test whether the same declared geometry survives new identities without moving its poles, boundaries or scoring rules.

Crosswalks establish compatibility and may expose a useful common representation. They are not empirical confirmation that ARA generated the established law. Structural recovery is stronger when the target relation was not used to choose the construction, but it still requires a non-guaranteed prediction before it becomes evidence beyond convergence. The universal bedrock proposal cannot be established by accumulating algebraic rewrites alone.

## 1.3 What the first paper proposed

The April 2026 paper, *The Geometry of Time: A Heuristic Framework for Classifying Non-Equilibrium Thermodynamic Systems* (La Franchi, 2026), began from a simple empirical observation: many cyclic systems spend distinguishable amounts of time accumulating a capacity and releasing it. It defined

$$\mathrm{ARA}_{\mathrm{duration}} = \frac{T_{\mathrm{release}}}{T_{\mathrm{accumulation}}}$$

and used that ratio to place examples from biology, geophysics and engineered or forced oscillators on a common temporal spectrum. The early taxonomy treated values near `1` as forced or pacemaker symmetry, proposed `\phi` as an attractor for free-running self-organising systems, and associated the extremes with singular or harmonic limits. It further proposed that distance from `\phi` could indicate forcing, dissipation or pathology.

That paper was intentionally broad and exploratory. The systems were not sampled randomly, their independence was not established, and several classifications developed during inspection. Its numerical clustering and simulations therefore preserve the historical origin of the hypothesis but are not treated here as confirmatory evidence for a universal constant.

## 1.4 What survived, what changed and why correction matters

The durable insight is relational rather than constant-specific: a declared identity can be studied through the asymmetry of two opposed contributions, and the result depends on the chosen boundary, direction, scale and time window. The accumulation/release duration ratio remains a valid instrument when those are the measured quantities.

The framework has nevertheless changed substantially:

- the wave or sphere is now the primary object; a scalar ARA value is one cut through it;
- `0–2` is a reversible normalized coordinate rather than a universal catalogue of physical substances;
- `1.0` is the selected parent ridge or core, not automatically a forced pacemaker;
- endpoint meanings depend on declared orientation, and a coordinate endpoint does not by itself prove physical failure;
- TE-ARA separates normalized allocation from native physical magnitude;
- rungs, child projection, Information³, Di-ARA and ARA⁹ make the proposed recursion explicit;
- standalone `\phi` has been demoted from universal attractor to one candidate within a broader irrationality programme; and
- crosswalks, structural recoveries, empirical predictions and physical hypotheses now receive different evidential labels.

These corrections are part of the result. A framework claiming to be load-bearing across domains must preserve its failures and narrow rules that do not survive frozen tests. Version 2 therefore supersedes the first paper's universal Phi interpretation without erasing its provenance or the observations that motivated the wider geometry.

The complete migration table is maintained in `ARA_PAPER_V2_CLAIM_MIGRATION.md`.

# 2. Reading rules and claim classes

## 2.1 Definition

A definition fixes the object or vocabulary used by the framework. For example, declaring an ARA identity to include a boundary, rung, time window and orientation establishes what will be measured. A definition can be internally inconsistent or empirically unhelpful, but it is not itself evidence that nature realizes the object universally.

## 2.2 Exact mathematical consequence

An exact consequence follows deductively after the definitions are accepted. The mirror map

$$M(x) = 2 - x$$

is exact on the declared `0–2` chart, as are (M(M(x))=x) and (M(1)=1). These statements prove properties of the coordinate. They do not prove a physical phase flip, a universal octave law or a hidden mechanism in data.

## 2.3 Typed crosswalk

A typed crosswalk is an exact or faithful translation of an established relation into declared ARA coordinates. It must preserve the source law's quantities, assumptions and predictions. It demonstrates that the two descriptions are compatible; it does not count as independent empirical evidence for ARA and must not be described as ARA having discovered the established law.

## 2.4 Frozen empirical result

An empirical result requires a declared object, outcome, operator, controls and verdict rule fixed before the relevant targets are examined. Where practical, the prediction artifact is hashed before reveal and the calculation is independently reproduced. Results are reported as **supported**, **not supported**, **inconclusive** or **descriptive only** under the named protocol. Passing a frozen test supports that specific rule on that source; it does not automatically establish the universal framework.

## 2.5 Conditional physical hypothesis

A conditional physical hypothesis adds a claim about what the coordinate represents in nature. Examples include universal wave/sphere recursion, physical continuation through an endpoint, resonance death, or a Space/Time assignment to the poles. Such a statement can be motivated by exact geometry and domain crosswalks, but it requires its own observable predictions and falsifiers.

## 2.6 Provenance and temporal priority

Temporal priority is established by dated files, frozen protocols, hashes, test identifiers and the provenance ledger—not by later recollection. The paper distinguishes a rule stated before a result, a post-result interpretation, a later correction and an independent replication. Earlier wording is retained in the archive even when superseded, so that neither successes nor failures can be rewritten retrospectively.

## 2.7 Typed blindness and geometry-first construction

This paper does not use *blind* as an unqualified synonym for independent discovery. It distinguishes at least four methodological conditions:

1. **Human theory-naive:** the human author did not use the domain's established formalism to choose the proposed ARA architecture.
2. **ARA-first or geometry-first:** identities, boundaries, rungs, phase assignments and relations were declared in ARA language before conventional domain labels were attached.
3. **Target-hidden:** the outcome being predicted was sealed until a prediction artifact and its hash had been saved.
4. **Untouched-source:** the source archive had not been inspected during construction of the rule.

These conditions are not interchangeable. In the quantum programme, the human author's construction became explicitly bottom-up and ARA-first at the Q16 methodological restart on 25 July 2026. The AI collaborator nevertheless possessed general quantum knowledge and handled much of the translation into established physics. The human-AI pair was therefore not fully theory-blind. Each quantum test is labelled by the blindness it actually achieved rather than receiving one blanket description.

# 3. The core ARA identity

## 3.1 The whole precedes the cut

An ARA identity is a declared coupled whole

$$\Omega = (\mathcal{B}, k, \Delta t, \hat{n}, \mathcal{R}),$$

where $\mathcal{B}$ is the measurement boundary, $k$ is the selected rung or scale, $\Delta t$ is the time window, $\hat{n}$ is the orientation of the cut and $\mathcal{R}$ is the set of retained internal or external relations. The framework models $\Omega$ as a wave or sphere with two opposed source directions.

One scalar observation is a diameter projection

$$P_{\hat{n}}(\Omega) = x_{\hat{n}} \in [0, 2].$$

The scalar is therefore not the sphere. It is a compressed answer to one relational question. A different axis, boundary, rung or window can produce a different reading without implying that the underlying whole has contradicted itself. Multiple justified cuts can be retained to reconstruct more of the whole.

## 3.2 The oriented `0–2` diameter

The oriented diameter has endpoints

$$P_0 = 0, \qquad P_2 = 2,$$

with ridge-centred coordinate

$$s = x - 1 \in [-1, 1].$$

The numbers are relational landmarks, not universal physical units. A calculation must still specify the instrument that maps the observed quantities to this interval. Duration normalization, signed flux, reciprocal/log scaling and shared-boundary cancellation are different instruments for different data types; they are introduced in Section 7.

Orientation reversal is

$$M(x) = 2 - x.$$

It preserves distance from the ridge:

$$|M(x) - 1| = |x - 1|.$$

This is the mathematical reason ARA can be read reversibly. Reversing the chart exchanges pole labels but does not alter the measured state.

## 3.3 Phase A, Phase B and reversible orientation

After orientation is fixed, ARA represents two counter-traversing lineages:

$$\text{Phase A} : 0 \to 2, \qquad \text{Phase B} : 2 \to 0.$$

The same endpoint is therefore the source of one wave and the terminal pole of the other:

| Pole | Wave sourced there | Wave completing there |
|---|---|---|
| `0` | Phase A | Phase B |
| `2` | Phase B | Phase A |

Phase A and Phase B name the declared lineages. *Accumulation* and *release* name what a lineage is doing relative to a particular boundary. A wave can release from a parent while accumulating into a child or neighbour. Phase A is therefore not universally identical to accumulation, nor Phase B to release.

Domain labels such as Space/Connection and Time/Traversal may be attached when a test justifies them. The geometry remains symmetrical, so those labels may be reversed for clarity provided the orientation is declared before scoring.

## 3.4 The `1.0` ridge/core

At

$$x = 1,$$

the two declared contributions meet equally and oppositely on the selected parent cut. In a signed observable their retained parent effects cancel; in the sphere model the diameter compresses to its central ridge or core.

The parent value alone does not reveal its microscopic cause. The same `1.0` reading can contain:

- genuinely balanced same-rung contributions;
- asymmetric children whose signed effects cancel after projection;
- a child endpoint projected upward into one parent phase;
- perpendicular activity invisible to this diameter; or
- a contextual contribution omitted from the selected grain.

Accordingly, `1.0` is not automatically stillness, resonance or externally forced symmetry. Those are typed physical interpretations requiring additional cuts or observables.

## 3.5 Endpoint singularities and the boundary between geometry and dynamics

ARA calls `0` and `2` the singularity poles of the selected identity because they are the endpoints at which one declared lineage is born and the counter-lineage completes. A proposed dynamical handover can be written

$$2_A \longrightarrow 0_B.$$

This seam crossing is not the same operation as the static chart reversal $x \mapsto 2 - x$. The first is a hypothesis about evolution through a boundary; the second is exact relabelling of one state.

The term *singularity* is therefore identity-relative in ARA. It does not automatically mean an infinite curvature, divergent field or breakdown of an established physical equation. Domain-specific singularity claims must be tested in that domain.

Fractal projection also separates parent and child meanings. A child can reach one of its own endpoints and contribute to a parent ridge without turning the parent's `1.0` coordinate into a same-rung endpoint. Likewise, physical resonance death—the complete dominance of one connection-like phase with no internally available reopening flow—is retained as a conditional hypothesis, not inferred from the normalized endpoint alone.

# 4. TE-ARA, activity and contextual participation

## 4.1 Native total `2`

TE-ARA is the total-allocation bookkeeping view of a selected ARA identity. The label was introduced because this same complete-identity account appeared often enough across decompositions, parent/child projections and contextual-coupling audits that leaving it unnamed caused ambiguity. It does not add a new object, axis, law or geometry to ARA. Once the boundary and grain are declared, the complete normalized identity has total

$$\boxed{\mathrm{TE\text{-}ARA}(\Omega) = 2.}$$

For a pure two-pole partition,

$$t_A + t_B = 2.$$

This closure contains two related but different counts. Phase A and Phase B are each represented by one canonical whole-phase contribution, so their symmetric meeting is

$$t_A = t_B = 1, \qquad 1_A + 1_B = 2.$$

At the same time, each counter-traversing wave has enough directional capacity to span the full `0-2` diameter:

$$C_A^{\mathrm{path}} = 2, \qquad C_B^{\mathrm{path}} = 2, \qquad C_{\mathrm{gross}}^{\mathrm{path}} = 4.$$

The gross `4` and the TE-ARA `2` are not competing totals. The first counts both complete directional paths separately. The second measures how much of those opposed paths can be expressed together inside one identity at one declared slice. Because advance by one wave opposes or displaces the other on the shared diameter, the contemporaneous pure allocation is constrained to

$$0 \leq t_A, t_B \leq 2, \qquad t_A + t_B = 2,$$

ranging from `2+0` at one extreme through `1+1` at the ridge to `0+2` at the other. In the framework's original energy language, the two waves possess a maximum gross traversal capacity of `4`, but opposition limits the realizable identity budget to `2`. Unless a domain supplies a common physical energy measure, *capacity*, *path budget* and *allocation* are the precise terms; this statement alone is not a claim of four joules reducing to two joules.

For an observed identity with named contextual participation,

$$\boxed{t_A + t_B + \sum_j t_{J_j} + O_{\mathrm{share}} = 2.}$$

The equation is an accounting closure at one declared perspective. Moving to a child, neighbour or parent creates a new identity with its own normalized total `2`; it does not divide a single universal budget among every scale in nature.

## 4.2 Geometric breakdown of the TE-ARA total

The TE-ARA account is generated directly by the diameter geometry; it is not an independent conservation rule placed on top of ARA. Let the selected oriented diameter be

$$\Gamma_\Omega = [P_0, P_2] = [0, 2].$$

With Phase A launched from $P_0$ toward $P_2$ and Phase B launched from $P_2$ toward $P_0$, the two paths occupy the same line in opposite directions:

$$\begin{array}{lccccc} \text{Phase A:} & P_0 & \longrightarrow & x & \longrightarrow & P_2 \\ \text{Phase B:} & P_0 & \longleftarrow & x & \longleftarrow & P_2. \end{array}$$

Each full source-to-terminal path has length `2`, which produces the gross path count

$$|P_2 - P_0|_A + |P_0 - P_2|_B = 2 + 2 = 4.$$

At one slice $x$, however, TE-ARA does not add both already-traced full paths. It reads the complementary capacities remaining from the two source orientations. Under this declared labelling,

$$\boxed{t_A(x) = 2 - x, \qquad t_B(x) = x, \qquad t_A(x) + t_B(x) = 2.}$$

The three principal cuts are therefore:

| Diameter position | Phase A allocation | Phase B allocation | Geometric reading |
|---|---|---|---|
| $x = 0$ | 2 | 0 | Phase A source / Phase B terminal |
| $x = 1$ | 1 | 1 | equal opposed ridge |
| $x = 2$ | 0 | 2 | Phase A terminal / Phase B source |

Reversing the orientation exchanges A and B and applies $x \mapsto 2 - x$; it does not change the total. The `2` is thus the length of one shared diameter partitioned complementarily, whereas the `4` is obtained only by tracing the two complete directed journeys separately.

For an observed identity whose direct A/B pair occupies only the subtotal

$$T_{AB} = t_A + t_B = 2 - \sum_j t_{J_j} - O_{\text{share}} > 0,$$

the same geometry is retained by normalizing within that pair:

$$x_{A/B} = 2\frac{t_B}{T_{AB}}, \qquad t_A = \frac{T_{AB}}{2}(2 - x_{A/B}), \qquad t_B = \frac{T_{AB}}{2} x_{A/B}.$$

This is the explicit bridge between ARA and TE-ARA: ARA supplies the location and orientation on the diameter; TE-ARA decomposes the identity's total allocation among the two directly expressed waves, named couplings and typed `Other`. If $T_{AB} = 0$, the A/B coordinate is undefined rather than zero.

The sphere interpretation adds perspectives, not extra energy. Rotating the diameter supplies other cuts through the same identity. Those cuts may expose perpendicular Di-ARA structure or previously hidden children, but they must not be summed as though every view were an independent TE-ARA budget. A child or parent sphere receives its own total `2` only after it is declared as a different measurement identity.

## 4.3 Allocation is not automatically physical energy

The word *energy* was used broadly during the framework's development to mean the capacity that moves through or deforms the geometry. In the formal account, TE-ARA components are normalized relational allocations. They are not automatically joules, power, probability, mass or information entropy.

A physical-energy reading is permitted only when all components are derived from a common energy budget and the conversion is shown. Otherwise the safer terms are *participation*, *transfer*, *coupling*, *accumulation* and *release*. This distinction prevents a conserved normalized total from being mistaken for a new conservation law in the underlying domain.

## 4.4 Native activity and magnitude

Normalization deliberately discards absolute scale. The framework therefore retains a separate native activity or magnitude

$$\Lambda_\Omega \geq 0.$$

Two identities can occupy the same ARA coordinate while one is microscopically weak and the other physically dominant. If the TE-ARA components partition one additive physical quantity, their dimensional contributions may be reconstructed as

$$Q_i = \frac{t_i}{2}\Lambda_\Omega, \qquad \sum_i Q_i = \Lambda_\Omega.$$

That reconstruction is valid only for a genuinely shared additive unit. In other applications $\Lambda$ can be event count, variance, flux magnitude or another domain-specific activity measure and must be named accordingly.

## 4.5 Typed `Other`: share, flow and difference

`Other` is not a third fundamental pole. It records what the selected A/B description has not yet resolved:

- $O_{\text{share}} \geq 0$ is an unresolved non-negative allocation inside the total-`2` account;
- $O_{\text{flow}} \in \mathbb{R}$ is a signed missing source, sink or boundary transfer; and
- $O_{\text{difference}} \in \mathbb{R}$ retains child asymmetry discarded during parent compression.

Once identified, an `Other` contribution becomes a named coupling such as drag, gravity, neighbouring flow or a lower-rung residue. If evidence instead shows that it was a missing part of the chosen Phase A/Phase B pair, the original decomposition must be corrected rather than protected by an expanding residual category.

# 5. Rungs, octaves and child-to-parent projection

## 5.1 Pure octave scaling

A pure octave is the framework's ideal scale step. A complete total-`2` identity at rung (k) generates the reference magnitude

$$\Lambda_{k+1} = 2\Lambda_k.$$

This is not a second geometry added to the within-identity coordinate. At rung $k$, the instantaneous diameter cut is

$$x_k(t) \in [0, 2],$$

and one idealized oriented cycle passes through the landmark sequence

$$1 \longrightarrow 2 \longrightarrow 1 \longrightarrow 0 \longrightarrow 1.$$

The part above the ridge is Phase A-leading and the part below it is Phase B-leading under the declared orientation. Peak, trough and ridge are gradients of the same closed wave; a sinusoid is not required. When the complete Phase A/Phase B cycle is treated as one child whole inside a larger closure, the same total-`2` structure appears cross-rung as $\Lambda_{k+1} = 2\Lambda_k$.

The factor `2` is therefore not an independently fitted constant: it is the scale expression of the same ARA closure that supplies the `0-2` diameter. The two equations are **distinct projections of one recursive wave/sphere**, not unrelated rulers. They must still not be substituted mechanically: $x_k = 2$ says that the current identity is at one pole, whereas $\Lambda_{k+1} = 2\Lambda_k$ says that a completed identity has been promoted into a pure next-rung relation. Intermediate identities can occupy gradient positions between pure octaves without changing the reference geometry.

In the generative hypothesis, two opposed waves and their retained relation mix into an identity that is itself a wave. That identity contains smaller child waves and participates in larger parent waves:

$$\cdots \longleftrightarrow \Omega_{k-1} \longleftrightarrow \Omega_k \longleftrightarrow \Omega_{k+1} \longleftrightarrow \cdots.$$

The recursion is mathematically repeatable without a built-in terminal rung. Whether physical reality realizes it indefinitely is part of the bedrock hypothesis; in empirical work the practical limit is the finest and largest identity the data can define.

Whether a particular physical hierarchy actually follows this exact doubling is empirical. Prime, quantum and other analyses provide tests and examples; the definition alone does not establish universality.

## 5.2 The directional half-weight rule

A child remains a complete total-`2` identity on its own rung. When one expressed child direction is read exactly one pure octave upward, it contributes half-weight to the selected parent phase. For a child coordinate $x_c \in [0, 2]$, the two alternative orientations are

$$c_{B|p} = \frac{x_c}{2}, \qquad c_{A|p} = \frac{2 - x_c}{2}.$$

These are alternative directional readings. They must not be added automatically, because doing so would count both chart orientations as simultaneous physical contributions.

## 5.3 Ridge children and full-phase child resonance

At the child ridge,

$$x_c = 1 \quad \Longrightarrow \quad c_{A|p} = c_{B|p} = 0.5.$$

A standard ridge child therefore supplies half of one selected parent phase. At the child endpoint, in the matching direction,

$$x_c = 2 \quad \Longrightarrow \quad c_{B|p} = 1.$$

This is the framework's **full-phase child resonance**: the child can occupy an entire Phase A or Phase B allowance of its parent. It must not be confused with the statement that the child's TE-ARA total equals `2`, because every complete child has total `2` regardless of its internal position. Full-phase resonance refers specifically to the child's coordinate reaching its `2.0` pole and projecting upward as `1`.

## 5.4 Parent compression and retained child asymmetry

Coarse-graining replaces a set of internal child relations with a smaller parent description. Opposed child deviations can cancel in the parent even while remaining strong below it. If

$$\delta_i = x_{c_i} - 1,$$

then a parent instrument may retain only a weighted aggregate such as

$$\Delta_p = \sum_i w_i \delta_i,$$

while discarding the individual $\delta_i$. A parent ridge $\Delta_p = 0$ therefore does not imply that every child is itself at `1.0`.

This is the formal version of the framework's flattening warning: measuring the whole can reveal the parent identity while hiding the children that produced it. Where later reconstruction matters, the discarded pattern must survive as named child readings, a typed difference residual or additional independent cuts.

## 5.5 Conditions for reversible reconstruction

Compression is exactly reversible only when either:

1. the internal relations were retained explicitly or in a sufficient typed residual; or
2. enough independent relations survive to determine the missing component under a declared and valid closure geometry.

The second route can be written

$$R_{\text{missing}} = F(R_1, R_2; G),$$

where (G) is the closure rule. Fractality alone does not restore information absent from both the measurement and the constraint. This boundary was demonstrated directly by the quantum missing-cell work: a complete $ARA^9$ representation was useful, but an arbitrary ninth value could not be recovered universally from the other eight under the proposed closure rule.

# 6. Information³ and the recursive relational hierarchy

## 6.1 Two distinguishable identities and their relation

ARA begins from one connected relational web $\Omega$, not from permanently isolated information packets. Local identities are declared projections of that web:

$$A = P_A(\Omega), \qquad B = P_B(\Omega), \qquad R_{AB} = P_{AB}(\Omega).$$

The act of measurement exposes or compresses (R_{AB}); it does not create all connection between (A) and (B). This is why ARA treats the relation as part of the information required to identify the pair.

## 6.2 Minimum relational closure

The minimum locally identifiable ARA object contains

$$\boxed{A, \quad B, \quad R_{AB}}.$$

This is the precise meaning of the framework's mnemonic "`1 + 1 = 3` information": two distinguishable identities expose a third informational component—their relation. It is not the ordinary arithmetic assertion $1 + 1 = 3$, and $R_{AB}$ is not a third pole on the ARA diameter. The poles remain `0` and `2`; the measured coordinate describes their relation.

One comparison is not always sufficient to lock a missing quantity. Under a declared geometry, two independent relations may constrain a third:

$$R_3 = F(R_1, R_2; G).$$

Pythagoras is one familiar Euclidean example of such closure, not the universal Information³ equation.

## 6.3 Recursive web formation

Once a relational unit closes, it can be compressed into one identity on the next rung:

$$(A, B, R_{AB}) \longrightarrow U_{AB}.$$

That new whole remains inside the same web and can enter further relations:

$$\text{relations} \to \text{closure} \to \text{identity} \to \text{new relations.}$$

Local triangular closure and extended lattice formation are therefore the same branch operation viewed at different scales. The name Information³ refers to this ternary relational foundation. A literal counting law such as $i + 1 = i^3$ is not assumed: cubic or faster combinatorial growth requires a separately declared convention retaining ordered relations, comparisons between relations or recursive depth.

## 6.4 Di-ARA as the first persistent perpendicular parent

When two complete ARA relations couple frequently and persistently enough to be treated as one recognizable parent, they form a Di-ARA:

$$D = (u, v) \in [0, 2]^2.$$

Di-ARA is not a second foundation. It is the first common decompression in which two ARA axes retain their independent orientations rather than being flattened immediately into one scalar. It can preserve state or direction across handovers that would be ambiguous on either diameter alone.

Independent chart reversals act as

$$M_u(u, v) = (2 - u, v), \qquad M_v(u, v) = (u, 2 - v),$$

and commute:

$$M_u M_v = M_v M_u.$$

The four Di-ARA sectors are therefore exact sign sectors of two centred ARA coordinates.

## 6.5 The `Ab`, `aB`, `bA`, `Ba` mixed branches

ARA uses letter order to record ownership or lineage and capitalization to record which component is expressed more strongly:

| Branch | Owning lineage | Stronger expressed component |
|---|---|---|
| `Ab` | A | A |
| `aB` | A | B |
| `bA` | B | A |
| `Ba` | B | B |

A canonical forward traversal is

$$Ab \to aB \to bA \to Ba \to Ab,$$

with the reverse traversal equally permitted when orientation is declared. The mixed casing is not decorative: `aB` and `Ba`, for example, can share a strong B component while belonging to different lineages and therefore carrying different handover histories.

## 6.6 ARA⁹ within and across lineages

ARA⁹ is the next decompressed neighbourhood. It retains three Information³ components—(A), (B) and their relation (R)—in a `3 x 3` structure.

Along one lineage, the columns can represent the centre, one step away and two steps away:

| | centre | step 1 | step 2 |
|---|---|---|---|
| $A$ | · | · | · |
| $B$ | · | · | · |
| $R$ | · | · | · |

Across two coupled lineages, the same geometry records all pairings of their three components:

| | $A_2$ | $B_2$ | $R_2$ |
|---|---|---|---|
| $A_1$ | · | · | · |
| $B_1$ | · | · | · |
| $R_1$ | · | · | · |

These are two directions through one relational web, not competing definitions of $ARA^9$. A complete neighbourhood can compress into a parent identity,

$$\mathrm{Compress}(\mathrm{ARA}^9) = \Omega_p,$$

but exact decompression still requires the reversal conditions in Section 5.5.

# 7. Mathematical instruments for one geometry

## 7.1 Positive two-part normalization

For positive measured parts $a, b > 0$, define

$$x = 2\frac{a}{a+b}.$$

Then

$$0 < x < 2, \qquad x = 1 \iff a = b, \qquad x(b,a) = 2 - x(a,b).$$

The unbounded raw ratio remains recoverable:

$$\frac{a}{b} = \frac{x}{2-x}.$$

This is the bounded form of the April duration ratio when (a) and (b) are the two declared phase durations. It is also applicable to other positive two-part measurements. Its algebraic validity does not prove that the chosen parts are the correct physical poles.

## 7.2 Signed boundary or flux instrument

Let $F_+ \geq 0$ and $F_- \geq 0$ be the magnitudes of positive and negative flow through a declared boundary. Define

$$\Lambda_F = F_+ + F_-, \qquad x_F = 2\frac{F_+}{F_+ + F_-}$$

when $\Lambda_F > 0$. The signed net flow is then

$$\boxed{F_{\text{net}} = \Lambda_F(x_F - 1) = F_+ - F_-.}$$

This factorization separates direction and balance from total activity. If $\Lambda_F = 0$, the boundary is inactive and $x_F$ is undefined; assigning `1.0` would confuse no measurement with balanced opposing flow.

## 7.3 Reciprocal/log instrument

For a positive reciprocal pair ((q,1/q)), the signed logarithm

$$\ell = \log q$$

changes sign when the pair is reversed. A bounded conversion $x = f(\ell)$ is ARA-compatible when its scale is published and

$$f(-\ell) = 2 - f(\ell).$$

Logarithmic coordinates are therefore natural for multiplicative ladders and scale ratios. They are an instrument acting on one typed dataset, not the definition of ARA and not evidence by themselves for octave doubling.

## 7.4 Shared-boundary cancellation

Suppose two children share an internal boundary. A transfer across it is release from one child and accumulation into the other. When the measurement boundary expands to enclose both children, that transfer no longer crosses the parent boundary and cancels from the parent account.

This is a standard boundary-conservation fact and an exact explanation for one kind of ARA compression. It also shows why changing scale changes the reading: internal activity can remain large while its net contribution to the parent boundary becomes zero.

## 7.5 Parent plus retained difference

One exact discrete representation of reversible compression retains a parent-like centre and the child difference. For integer children (a,b), let

$$m = b + \left\lfloor \frac{a-b}{2} \right\rfloor, \qquad d = a - b.$$

The inverse is

$$b = m - \left\lfloor \frac{d}{2} \right\rfloor, \qquad a = b + d.$$

Here (m) is the compressed central value and (d) preserves the signed asymmetry needed for exact reconstruction. This proves one implementation of the parent-plus-difference principle. It does not establish useful compression ratios, noise tolerance or cryptographic security.

## 7.6 Why these are instruments rather than one universal arithmetic formula

Duration ratios, signed flux, logarithmic scale, averaging, coherency and relational closure expose different quantities. ARA's common content is the declared two-pole identity, reversible orientation and retained relation—not one arithmetic operation forced onto every dataset.

Every instrument must therefore publish:

1. the identity and boundary;
2. the two pole observables;
3. the chart orientation;
4. the native activity or magnitude;
5. the rung and any child/parent projection;
6. what survives as a named coupling or typed `Other`; and
7. an outcome that is not guaranteed merely by the normalization.

An instrument that cannot be back-translated into the intended ARA question is a proxy test, even if it produces an attractive shape.

# 8. Test methodology

## 8.1 Declare the identity before scoring

Exploration is permitted to be open-ended. A result acquires evidential status only after its ARA object has been operationalized. Before scoring, the record must state the boundary, two opposed observables, orientation, rung, time window, retained relations and proposed outcome.

When a qualitative ARA statement is translated into mathematics, the original statement, proposed operator, plain-language back-translation and forbidden neighbouring proxies are recorded together. If the operator answers a different question, its result is classified as a proxy rather than negative evidence against the intended claim.

## 8.2 Freeze poles, orientation, scale and measurement cut

The poles and their direction must be fixed before looking at the target separation. Reversing the chart after inspection, moving the boundary, changing the child depth or selecting a more favourable time cut creates a new hypothesis. It may be explored, but it cannot rescue the frozen one.

Every visualization must distinguish measured coordinates from assigned landmarks or aesthetic guides. A line drawn at `1`, `\phi` or another value is not an observation unless the coordinate that generated it is stated.

## 8.3 Separate development, evaluation and untouched holdout

Development data may be used to calibrate extraction, choose among declared instruments and discover failure modes. Evaluation data tests the frozen choice. An untouched holdout or external archive is reserved for the claim of generalization.

When chronological prediction is intended, every transform—including centring, detrending and anomaly calculation—must be computed from information available at the forecast origin. Random train/test splitting does not establish forward predictability in a time series.

## 8.4 Require controls that can defeat the proposed constant or geometry

Controls are selected to answer the strongest plausible alternative explanation. Depending on the claim, they include:

- ridge, persistence, reversal and no-coupling rules;
- fitted identity-specific constants;
- rational, irrational and randomized competitors;
- shuffled labels or destroyed temporal order;
- domain-standard physical or statistical predictors; and
- synthetic data calibrated to the record's length and signal-to-noise ratio.

A universal constant must defeat fitted and neighbouring alternatives, not merely appear near a noisy estimate. A structural geometry must outperform simpler summaries when its claim is that the extra structure matters.

## 8.5 Preserve failures and post-result reinterpretations

The frozen claim verdict and the descriptive geometry verdict are reported separately. A prediction can fail while exposing an informative shape shared by target and controls; an attractive post-hoc shape cannot reverse the failed prediction.

Corrections are timestamped and old artifacts remain available. The vocabulary used in the repository is:

- **confirmed** for a preregistered result surviving controls and independent replication;
- **supported** for a positive controlled result not yet independently replicated;
- **suggestive** for a matching direction with substantial fences;
- **inconclusive** when the instrument cannot decide;
- **null** when an adequate instrument finds no effect;
- **not supported** when the frozen prediction fails cleanly; and
- **retracted** when a prior claim is subsequently killed.

## 8.6 Independent validation and reproducible artifacts

Every empirical section must identify the source, download date, data checksum or stable identifier, protocol, executable script, prediction artifact, raw result and validation report. Scripts should reproduce the registered result from a fresh clone, automatically fetching public data when licensing permits.

Independent validation means more than rerunning the same summary function. It should reconstruct the analyzed records from the source, verify partitions and eligibility, recompute the main metrics and check that the verdict follows from the frozen gates. A surprising result also requires split stability, clustered uncertainty where observations share origins, and a negative or randomized control.

## 8.7 ARA-first reveal order

Where a test asks whether ARA can recover a known scientific structure from the bottom up, the preferred order is:

1. anonymize or withhold conventional domain labels where possible;
2. declare the ARA object and measurement operator;
3. freeze thresholds, controls and partitions;
4. save the ARA result;
5. attach the conventional scientific labels and compare structures;
6. make a separate prospective prediction if the recovery itself could not fail independently.

This procedure tests whether geometry-first construction converges on meaningful established structure without pretending that convergence is the same as a blind discovery. Any prior exposure of the human author, AI collaborator or test designer must be stated.

# 9. Crosswalks to established science

## 9.1 Crosswalk rule: translation is not proof

Every crosswalk must show the established source relation beside the ARA representation, identify which steps are definitions or normalization, and state what information—if any—the ARA representation adds. Algebraic equivalence is classified as compatibility. It is not classified as a novel prediction, an independent empirical replication or proof of the bedrock hypothesis.

## 9.2 Structural recovery beyond relabelling

The stronger recovery question is whether predeclared ARA rules reconstruct a non-trivial feature of an established result without importing that feature as an input. Such a recovery must identify the available prior information, the degrees of freedom, plausible controls and the feature that could have come out differently. It remains distinct from a prospective prediction on untouched data.

## 9.3 Gauss and signed boundary closure

Gauss's electric law, here used in its modern SI integral form within the Maxwell system (Maxwell, 1865), is

$$\oint_{\partial V} \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{inside}}}{\varepsilon_0}.$$

Split the boundary flux into outward and inward magnitudes $\Phi_+ \geq 0$ and $\Phi_- \geq 0$, and define

$$\Lambda_\Phi = \Phi_+ + \Phi_-, \qquad x_\Phi = 2\frac{\Phi_+}{\Phi_+ + \Phi_-}.$$

Whenever $\Lambda_\Phi > 0$, Gauss's law becomes

$$\boxed{\Lambda_\Phi(x_\Phi - 1) = \Phi_+ - \Phi_- = \frac{Q_{\text{inside}}}{\varepsilon_0}.}$$

This is an exact reversible ARA factorization of signed boundary flux into activity and orientation. At $x_\Phi = 1$, equal non-zero inward and outward contributions cancel in the net Gauss reading while the boundary remains active. If $\Lambda_\Phi = 0$, the coordinate is undefined rather than a measured ridge.

The crosswalk does not modify or rederive Gauss's law. It shows that the ARA ridge can have a precise established meaning—signed boundary cancellation—and that the activity variable is required to distinguish balanced flux from an empty boundary. Whether the same decomposition supplies novel predictions about other identities is a separate empirical question.

## 9.4 Maxwell and perpendicular coupled fields

Maxwell's equations in vacuum-compatible SI form (Maxwell, 1865) are

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \qquad \nabla \cdot \mathbf{B} = 0,$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \qquad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$

They supply several distinct ARA-shaped relations. Gauss's magnetic law is a closed-boundary zero-net-flux identity. Faraday's law couples a changing magnetic field to circulating electric structure. The Ampère–Maxwell law couples magnetic circulation to conduction current and changing electric field. Charge continuity,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0,$$

is a literal local accumulation/boundary-release balance.

For a source-free plane wave,

$$\mathbf{E} \perp \mathbf{B} \perp \mathbf{k}, \qquad \mathbf{B} = \frac{1}{c} \hat{\mathbf{k}} \times \mathbf{E}.$$

At a fixed point the electric and magnetic amplitudes reach their peaks together; they are not quarter-cycle temporal anti-phases. The cyclic handover appears in the curl/time-change equations and in their mutually perpendicular spatial roles. Retaining electric and magnetic identities as separate axes is therefore a natural Di-ARA crosswalk, while flattening them immediately into one scalar discards direction.

The two field-energy densities are

$$u_E = \frac{\varepsilon_0 E^2}{2}, \qquad u_B = \frac{B^2}{2\mu_0}.$$

Since $E = cB$ in a vacuum plane wave, $u_E = u_B$. A TE-ARA energy allocation consequently gives

$$t_E = 2\frac{u_E}{u_E + u_B} = 1, \qquad t_B = 1.$$

This is an active equal-energy ridge. The third directed relation is the Poynting vector

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B},$$

which is perpendicular to both fields and records their joint energy-flow direction. In Information³ language, $\mathbf{E}$, $\mathbf{B}$ and $\mathbf{E} \times \mathbf{B}$ provide an exact example of two distinguishable components plus a relational direction that cannot be recovered by averaging their amplitudes.

Poynting's theorem completes the boundary account:

$$\frac{\partial u_{\mathrm{EM}}}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}.$$

It separates changing stored field energy, boundary transport and transfer to matter. These are exact Maxwell crosswalks. The public-data Maxwell tests validated the ARA implementation and scale-axis representation, but because their targets follow from Maxwell's equations they do not independently establish a universal ARA mechanism.

## 9.5 Newtonian and relativistic scale relations

For two interacting Newtonian bodies, the action–reaction relation traced to Newton's *Principia* (Newton, 1687) is

$$\mathbf{F}_{A\leftarrow B} = -\mathbf{F}_{B\leftarrow A}.$$

The forces act on different bodies and do not disappear. They cancel only after the measurement boundary encloses the pair:

$$\frac{d}{dt}(\mathbf{p}_A + \mathbf{p}_B) = \mathbf{F}_{A\leftarrow B} + \mathbf{F}_{B\leftarrow A} = 0$$

when no external force is present. Equal non-zero magnitudes therefore form an active ARA ridge for the internal-force account of the enclosing identity. Measuring either body or their relative separation produces a different lawful reading.

Newton's second law has the same signed-boundary factorization along a chosen axis. If $F_+$ and $F_-$ are opposed force magnitudes,

$$\Sigma_F = F_+ + F_-, \qquad x_F = 2\frac{F_+}{F_+ + F_-},$$

then

$$ma_{\parallel} = F_+ - F_- = \Sigma_F(x_F - 1).$$

Newton's first law occupies the zero-net-force condition, but $\Sigma_F$ distinguishes an inactive zero-force case from an active internal ridge. This is why ARA must retain magnitude as well as normalized position.

General relativity (Einstein, 1916) supplies an established scale reduction rather than a literal octave proof. Under the stationary weak-field, slow-motion and negligible-pressure assumptions,

$$g_{\mu\nu} \longrightarrow \Phi \longrightarrow \mathbf{g} = -\nabla\Phi \longrightarrow m\ddot{\mathbf{r}} = m\mathbf{g},$$

and Einstein's source equation reduces to

$$\nabla^2\Phi = 4\pi G\rho.$$

ARA interprets this as a typed cross-scale compression from spacetime geometry to potential, field tendency and matter response. The reduction is established physics; the stronger ontology that Space and Time are parent ARA waves whose children are Matter and Field is a conditional ARA hypothesis, not a consequence of the weak-field limit.

The equivalence principle reinforces the perspective rule: a gravitational-force description and a local freely falling geometric description can represent the same physics from different frames. It does not by itself establish the ARA sphere or universal fractal recursion.

## 9.6 Hamiltonian allocation and Noether conservation

Using Hamilton's dynamical formulation (Hamilton, 1834), an isolated ideal harmonic oscillator has

$$H = \frac{p^2}{2m} + \frac{kq^2}{2} = K + V.$$

With

$$Q = \sqrt{k}\, q, \qquad P = \frac{p}{\sqrt{m}},$$

the conserved-energy surface is the exact circle

$$Q^2 + P^2 = 2H.$$

The corresponding TE-ARA allocation is

$$t_V = 2\frac{V}{H}, \qquad t_K = 2\frac{K}{H}, \qquad t_V + t_K = 2.$$

Orienting the diameter toward kinetic expression gives

$$x_H = 2\frac{K}{H} = 2\frac{P^2}{Q^2 + P^2}.$$

Thus `0`, `1` and `2` mean all potential energy, equal kinetic/potential energy and all kinetic energy for this instrument. Hamilton's equations,

$$\dot{Q} = \omega P, \qquad \dot{P} = -\omega Q,$$

generate a continuous four-quadrant rotation. The energy diameter folds sign and direction, so a complete dynamical state must retain the quadrant or an equivalent derivative. This is an exact example of the sphere/circle preceding its scalar cut.

Noether's theorem (Noether, 1918; English translation 2005) supplies the essential conservation fence. TE-ARA closure

$$\sum_c t_c^{(\mathcal{P})} = 2$$

is true by normalization at perspective $\mathcal{P}$. A Noether charge instead satisfies

$$\frac{dQ_{\text{physical}}}{dt} = 0$$

because the action is invariant under a continuous transformation. Energy, linear momentum and angular momentum conservation arise from time, spatial and rotational symmetry respectively. An open subsystem can be normalized to total `2` while gaining or losing native energy; conversely, a state can move continuously around its ARA allocation while a Noether charge remains fixed. TE-ARA is therefore not a replacement for Noether conservation.

The virial theorem (Clausius, 1870) provides a further scale-spanning crosswalk. For bound inverse-distance potentials,

$$2\langle T\rangle = |\langle V\rangle|.$$

A weighted ARA comparison of $2\langle T\rangle$ and $|\langle V\rangle|$ lies exactly at `1.0` from planetary orbits to ideal nonrelativistic hydrogen. The raw energy allocation remains asymmetric at $2/3$ and $4/3$; the ridge belongs to the weighted virial identity, not to every energy cut. This demonstrates typed cross-scale consistency among systems already governed by the same inverse-distance structure, not independent proof that ARA generated the virial theorem.

## 9.7 Crosswalk summary and evidential limits

| Established relation | Exact ARA instrument or structure | Meaning of the ridge | Evidential classification |
|---|---|---|---|
| Gauss electric law | Signed boundary activity $\Lambda_\Phi(x_\Phi - 1)$ | Equal inward/outward flux contributions | Exact re-coordinateization |
| Newtonian internal pair | Opposed force magnitudes plus activity | Equal anti-directed internal forces at the enclosing boundary | Exact pair cancellation; universal ARA reading proposed |
| Vacuum Maxwell wave | Perpendicular E/B axes, equal energy allocation and cross-product flow | Equal electric and magnetic field energy | Exact electromagnetic identity and Di-ARA crosswalk |
| Harmonic Hamiltonian | $Q^2 + P^2 = 2H, t_V + t_K = 2$ | Equal potential and kinetic energy | Exact circle/allocation crosswalk |
| Inverse-distance virial theorem | Weighted $2\langle T\rangle$ versus $|\langle V\rangle|$ | Virial equality | Exact theorem crosswalk across typed scales |
| GR weak-field limit | Geometry to potential to Newtonian response | No universal `1.0` asserted | Established approximation; ARA rung interpretation conditional |

These examples show that ARA can provide a consistent relational language across several foundational equations without changing their established predictions. They also demonstrate why the identity declaration is indispensable: `1.0` can mean signed-flux cancellation, reciprocal force exchange, equal field energies, equal oscillator energies or a weighted virial equality.

The crosswalk programme would become evidence for a generative bedrock only if the same predeclared composition rule predicted non-guaranteed structure that the established equations had not already supplied to the calculation, and if that rule transferred unchanged across independent systems. Until then, the crosswalks establish mathematical compatibility, disciplined translation and candidate points for stronger tests.

# 10. Empirical domain tests

Each case study must state the object, frozen prediction, controls, verdict, limitations and reproduction path.

## 10.1 Pendulum and coupled-child decompositions

The public dynamicslab MultiArm-Pendulum archive (Kaheman et al., 2023; versioned dataset) provides a useful mechanical case because the same declared hierarchy can be inspected under free and externally driven motion. For the free triple pendulum, angle was mapped to an ARA position by

$$x_i(t) = 1 + \frac{\theta_i(t) - \theta_{i,\mathrm{rest}}}{\pi},$$

with `1.0` declared as the hanging rest ridge and the folded `0/2` endpoint as the upright pole. None of the free runs reached that pole, so these data cannot test a singularity crossing. They do test how three coupled rungs move around a shared ridge.

After replacing an over-sensitive turn detector with a prominence-and-spacing detector, the lowest arm led the most turns and held the longest single leadership block in all three runs. The apparent orderly split between the two upper arms did not survive: their lead shares were erratic, and earlier long blocks were partly micro-extrema artifacts. A two-mode reconstruction retained `99.4%` of the motion, but a period-ago baseline tied or beat several fitted forecasts. The more specific causal test was stronger: using only the histories of arms 1 and 2 to predict the held-out deepest arm gave correlation near `0.99` from nowcast through two seconds and beat that arm's own self-autoregression at two seconds (`0.988` versus `0.959`). This supports a bounded child/parent reading in the locked, quasi-periodic regime: the deepest visible child is strongly constrained by the shallower coupled structure.

The driven record supplied a distinct control. The free system's shared period was approximately `1.33 s`; under cart forcing every arm locked near the imposed `1.56 s` period, and the bottom arm's clock-likeness rose from `0.77` to `0.95`. The common fundamental therefore transferred down the chain, while harmonic phase and gain became progressively identity-specific. In ARA language, the fundamental marks the shared parent cadence and the harmonic tail retains which rung is being observed. In ordinary mechanics, this is entrainment and mode structure; ARA's contribution is the declared hierarchy and the separation of common parent timing from child-specific residuals.

The limitations are substantial. The decomposition uses standard normal-mode, spectral and causal-regression instruments; the free runs are tame; only one driven triple record exists; and neither record contains a complete pole crossing. Moreover, three frozen pendulum searches for a universal Phi placement failed: dimensional handover passed only `1/4` families, ordered recurrence passed `0/5`, and the state-sphere breathing coordinate passed `0/4`. A four-child allocation test was mixed (`2/4`) and did not establish a universal constant. The pendulum evidence therefore supports coupled-rung organization and parent/child predictability, not a universal Phi clock or new pendulum mechanics.

Primary reproduction paths are `analysis/pendulum_scripts/PENDULUM_ARA_RESULT.md`, `analysis/pendulum_scripts/PENDULUM_DRIVEN_ARA_RESULT.md` and the four frozen `PHI_*_RESULT_2026-07-30.md` reports in the same directory.

## 10.2 Quantum: bottom-up ARA-first reconstruction, ARA⁹ and falsification

### 10.2.1 The methodological reset

The quantum programme is important because, after a recorded correction in method, its conceptual direction was deliberately reversed. It did not begin by taking a finished quantum description and assigning ARA names to its components. From Q16 onward, the human author instead proposed parents, children, perpendicular cuts, rungs, flips and retained relations in ARA language; the raw or reconstructed quantum quantities were then used to test whether that geometry was present. Conventional quantum terminology was attached after the ARA construction wherever the protocol allowed it.

This is a bottom-up structural-recovery programme, but it is not described here as globally blind. Earlier quantum results and archive outcomes were already known, and the AI collaborator knew the relevant physics. The evidential question is narrower: did a predeclared ARA construction recover or predict a non-guaranteed structure, under controls capable of defeating it?

### 10.2.2 Q16: ARA-first raw four-child restart

Q16 declared two complete ARA parents and four ordered children before scoring four public raw-current archives from the Figshare Bell-state tomography deposit (Madzik and Asaad, 2021). The source files were renamed in immutable file-ID order; all nine unnamed measurement settings and five current segments were retained. The ARA stage did not use Bell-state names, Pauli axes, density matrices, Ramsey filters or Hahn filters to construct the geometry.

The frozen test passed `8/8` gates. Its four-child balanced accuracy was `0.8875`, compared with a label-shuffle 99th percentile of `0.487625`, and the two parent holdout cosines were `0.877841` and `0.829311`. Only after the ARA result had been saved were the strongest recovered directions translated to the established Bell-family relation.

The result is classified as a strong ARA-first raw-data crosswalk and structural recovery, not a blind discovery of Bell structure. Archive outcomes and Q4–Q15 were already known; preparation labels were correlated with separate acquisition archives; and Q16 did not outperform established quantum tomography. Its value is that a geometry declared in the framework's own language landed on a physically meaningful established relation while retaining the ability to fail.

### 10.2.3 Q24 and Q25: complete relation object versus hidden-cell prediction

Q24 identified the connected `3 x 3` relation tensor as an ARA⁹ neighbourhood: three relations on each side and the nine pairwise cuts between them. This is an exact and useful crosswalk of the complete object, not evidence that any eight cells uniquely determine the ninth.

Q25 tested that stronger claim on a different public quantum source: the data accompanying the nondestructive Bell-state measurement of two distant atomic qubits (Welte et al., 2021; data deposit). Each of `81` missing-cell predictions was written and hashed before the sealed target was read. The frozen ARA⁹ sphere-closure predictor produced real signal: on the primary external matrices its MAE was `0.12394`, beating the ridge control (`0.18616`) and the other-eight-cell mean (`0.21630`). It nevertheless lost to the physical-positivity midpoint (`0.08687`) and passed only `7/12` frozen gates. The primary claim was therefore **not supported**.

This pair of results is methodologically load-bearing. ARA[9] can faithfully organize the full relation object, yet its proposed local closure rule did not supply enough information to reconstruct an arbitrary hidden quantum cut. The bottom-up programme is retained; the failed universal reconstruction rule is not.

#### 10.2.4 Replication boundary

A later ARA-first boundary-child flow rule (Q33B) was supported inside its opened simulator. The unchanged rule was then frozen before an untouched cross-archive test (Q34) and did not replicate: its positive fraction was `54.21%`, below the frozen `55%` gate, with additional paired and bootstrap failures. This distinguishes an informative geometry inside one realization from a domain-invariant law.

Taken together, the quantum record supports a serious but bounded statement: ARA-first decomposition has repeatedly organized non-trivial quantum relations and generated falsifiable tests, including failures. It does not yet establish that quantum mechanics is generated by ARA, reveal a new quantum state, replace standard quantum theory or prove the universal fractal-sphere hypothesis.

### 10.3 Prime child/parent structure and the failed constant-step predictor

Prime numbers were used as an unusually severe test of the framework's child/parent language. The test object was not a continuous physical wave but the ordered survival of integer candidates as lower prime-factor gates are added. This produced several exact crosswalks. Clearing every prime divisor through the square-root boundary recovered all `46,903` primes in a fresh million-integer interval; reversible residue pairs allowed one member of each pair to reconstruct the other with zero errors across `92,160` held-out residues and `40/40` validation checks. These are exact structural correspondences to trial division and wheel symmetry. They are not new primality theorems or faster algorithms.

The strongest prospective locator was PN26. From `6,000` fresh anchors across three scales, the first complete lower-parent quiet state was the actual next prime `93.983%` of the time; the first two states contained it `99.650%` of the time and the first three `99.967%` of the time. Only one target appeared at rank four and one at rank five. Independent reconstruction passed `16/16` checks. The frozen all-gates claim was still only partial: the three-state list beat a `p <= 29` wheel by `37.60` percentage points rather than the required `50`, and each parent retained between `780` and `48,817` lower factor gates. The impressive ranked compression therefore remains a child-heavy partial sieve, not a constant-cost decoder.

The proposed cheap lift made that boundary explicit. The frozen one-shot rule

$$\widehat{P} = N + a + 2b + 1$$

hit a prime on `2,703` of `30,000` fresh odd anchors (`9.010%`), versus `8.777%` for equal use of the six allowed offsets. A matched permutation gave `p=0.0144`, suggestive but above the prospectively required `p<0.01`. The fallback child erased much of the benefit, and every even anchor remained even. The rule retained a small amount of local divisibility information; it did not solve prime location in three arithmetic steps.

Other frozen failures prevent a geometric rescue. A same-scale golden-cross carrier had AUC `0.497180`, a bootstrap interval spanning chance and circular-shift `p` values of `0.9455` and `0.9883`; all five registered gates failed. Dynamic child orientation improved the difficult unresolved-composite AUC from `0.5301` to `0.5663`, but its frozen permutation result was `p=0.06199` and therefore remained suggestive rather than replicated. The prime programme was consequently parked with a two-output verdict: exact and transferable arithmetic crosswalks are supported, while a cheap ARA-specific prime calculator, universal Phi carrier and asymptotic advantage are not.

The consolidated audit and reproduction map is `analysis/primes/PRIME_THREAD_CAPSTONE_AND_CLOSURE_2026-07-21.md`. The principal frozen reports are `PN26_DOMINANT_PARENT_RIDGE_LOCATOR_REPORT.md`, `PN27_EXACT_FIT_CHILD_LIFT_REPORT.md`, `PN30_DYNAMIC_RELATIONAL_FLIP_REPORT.md` and `PN35_SAME_SCALE_GOLDEN_CROSS_REPORT_2026-07-22.md`.

## 10.4 LLM hierarchy, closure and parent/child residuals

The Pythia checkpoint family (Biderman et al., 2023) and its public massive-activation archive (Sun et al., 2024; derived activation dataset) supplied a non-physical but highly structured hierarchy: model as parent, transformer layer as child and training checkpoint as ordered time rung. T338 froze a simple forecast in which the recent motion of one layer and the recent median motion of its model parent contributed equally:

$$\widehat{C}_{t+1} = \tfrac{1}{2} C_{\mathrm{total},t} + \tfrac{1}{2} P_t.$$

Across `335` clean checkpoint files, nine model sizes, `6,648` reduced layer records and `70,752` forecast rows, the result was mixed/partial. On the untouched 2.8B, 6.9B and 12B models it achieved `77.69%` direction accuracy, `1.36` percentage points above the local-flow direction, and log-MAE `0.118541` versus persistence `0.144781`. It also narrowly beat child-only (`0.119794`) and parent-only (`0.119962`) flow. It lost amplitude accuracy to local linear trend (`0.115799`), so the all-gates claim failed. Independent reconstruction passed `15/15` checks.

T339 then tested the scale wording more literally. A parent at its own rung has weight `1`, while a direct child projected upward has weight `0.5`. Applied naively as

$$P_t + \tfrac{1}{2} C_{\mathrm{total},t},$$

the rule double-counted the parent's common motion because the measured layer flow was not a pure child residual. Parent and total-layer flows correlated about `0.94`, and the literal rule's holdout MAE worsened to `0.146464`. Its frozen verdict was **not supported in this form**.

The post-result decomposition exposed the exact bookkeeping correction. If

$$C_{\mathrm{specific}} = C_{\mathrm{total}} - P,$$

then the intended parent-plus-half-child projection is

$$P + \tfrac{1}{2} C_{\mathrm{specific}} = P + \tfrac{1}{2}(C_{\mathrm{total}} - P) = \tfrac{1}{2} P + \tfrac{1}{2} C_{\mathrm{total}},$$

which is algebraically the original T338 operator. This is a useful framework correction rather than a new positive result: a child measurement must be stripped of inherited parent motion before receiving an additional parent contribution. Otherwise a true scale rule can be implemented incorrectly by flattening containment into addition.

The result does not show that language-model training is generated by ARA, and only three large model sizes formed the holdout. It does show that a predeclared parent/child coordinate carried modest directional information and that the framework's scale language made a falsifiable distinction between total child motion and child-specific residue. Reproduction is documented in `analysis/llm_training_dynamics/T338_LLM_TRAINING_DI_ARA_REPORT_2026-08-03.md` and `T339_LLM_TRAINING_PARENT_PLUS_CHILD_REPORT_2026-08-04.md`.

## 10.5 Solar, ENSO and complex-system prediction

The complex-system record is mixed and must be separated into raw forecasting performance, ARA-specific increment and architecture validity. The cleanest solar result used the WDC-SILSO monthly total sunspot record (Royal Observatory of Belgium) across approximately twenty-five solar cycles as a stable-clock system. Its self-forecast correlation was `0.853` versus a cycle-ago baseline of `0.685` at one year and `0.752` versus `0.687` at eight years; it met the cycle-ago floor near eleven years. This is strong raw predictability in a regular oscillator, but much of that predictability belongs to the Sun's cycle itself. The ARA-specific question is only whether the within-cycle state contributes beyond the cycle-ago clock.

For ENSO, the strongest audited strict-causal point forecast in the earlier TheFormula programme combined shape and reservoir magnitude using public Niño 3.4 anomaly and tropical-Pacific warm-water-volume records (NOAA PSL; NOAA PMEL). At a true six-month lead it reached correlation `0.506` versus persistence `0.410`, predicted-change correlation `0.458`, direction `0.629`, and MAE `0.615` versus `0.653`. Other exploratory models produced larger-looking improvements that did not survive shorter-window replication, strong lag controls or later leak audits. These outcomes motivate the accumulation/feeder/release architecture, but they do not uniquely identify ARA as the cause of the skill.

The most recent ENSO pair, T336/T337, is more important methodologically than numerically. Codex translated temperature and recharge variables into a perpendicular complex coordinate without first establishing that those measured identities were the intended ARA parents and children. The frozen implementation did not beat its controls: at six months the full handover's skill was below raw movement, and the direction model's balanced accuracy (`0.7353`) was below the raw-movement control despite a slightly higher AUC. Independent validators passed, so the computation is reproducible. The tests are nevertheless marked **architecture-invalid as tests of ARA's ENSO geometry**. They test the imposed `T+iR` encoding, not the framework architecture Dylan intended.

This invalidation is a load-bearing rule for future prediction work. The domain hierarchy must first declare the two ENSO parents, their children, three-generation lineage, inflow cut and same-rung external couplings. Only after that map is confirmed may it be translated, frozen and scored. The solar and ENSO programme therefore supplies useful causal forecasting benchmarks and a clear methodology correction, not yet an unknown complex-system solution.

The audited overview is `TheFormula/THEFORMULA_MAIN_THREAD_RESULTS_AUDIT_2026-07-11.md`; the solar case is reproduced under `TheFormula/12 - Heart ceiling, two-band ECG & solar flywheel (29-05-26)/`; the invalid ENSO translations and their preserved calculations are recorded in `TheFormula/T336_T337_ENSO_ARCHITECTURE_INVALIDATION_2026-08-03.md` and the adjacent T336/T337 report folders.

## 10.6 River, bubble and cross-domain Di-ARA coordinates

The most direct empirical test of Di-ARA used two explicitly different axes rather than a fixed irrational constant. In the public plunging-flow river archive (Li et al., 2024a; Dryad data), the first axis measured downstream contraction/expansion and the second measured reverse/forward turning. Their crossing generated the four typed sectors `Ba`, `Ab`, `bA` and `aB`. The reciprocal endpoint amplitude was fitted on calibration and then frozen; field order, intact elevation-rank lineage and thalweg specificity were tested against deterministic controls.

T335 passed every registered gate and `17/17` independent reconstruction checks. All four sectors were substantially occupied in untouched holdout (`27.05%`, `30.68%`, `22.22%`, `20.05%`). The field contraction and expansion endpoints were `0.894166` and `1.096260`, with product `0.980238`; the thalweg holdout endpoints were `0.914068` and `1.091394`, with product `0.997608`. Recorded downstream order beat all `1,000` deterministic order nulls in both evaluation and holdout (`p=0.000999` at the resolution floor), and breaking rank lineage increased holdout loss from `0.009982` to `0.128011`. The thalweg ranked `1/41` against matched depth-rank controls in holdout.

The public oscillating-flow bubble archive (Pandey et al., 2025) recovered the same four-sector reciprocal form but not the same dynamical strength. T334 retained `160` holdout transitions. Every sector was populated (`18.13%`, `26.88%`, `27.50%`, `27.50%`), and contraction/expansion endpoints `0.784672` and `1.223070` gave product `0.959709`. Intact bubble lineage strongly beat broken lineage: holdout loss `0.035214` versus `0.235742`. The predeclared chronological-order gate failed in holdout, so the full dynamical claim did not pass. Its correct verdict is support for the four-quadrant coordinate and identity-specific reciprocal breathing, not for one stable temporal operator.

These results also demote standalone Phi. The river Phi circle-train test selected persistence rather than Phi and gave downstream-order `p=0.550945`; the actual bubble-handover Phi seam also failed its frozen gates. A bubble circle-train test found a reproducible Phi return ranking but failed the parent, order and specificity gates. The cross-domain claim supported here is therefore structural: two declared ARA axes can produce a transferable four-sector relation with reciprocal contraction/expansion around a local ridge. The radial amplitude and amount of time ordering remain identity-specific; no universal Phi, `1/e`, `3/8` or other endpoint follows.

The primary reproduction reports are `analysis/hydraulics/T335_RIVER_IRRATIONALITY_DI_ARA_REPORT_2026-08-03.md` and `analysis/vertical_ara_bubbles/T334_BUBBLE_OCTAVE_RELATIVE_IRRATIONALITY_QUADRANT_REPORT_2026-08-03.md`. The fixed-Phi controls are T327/T319 in `analysis/hydraulics/` and T328/T329 in `analysis/vertical_ara_bubbles/`.

## 10.7 Empirical balance

The case studies do not all support the same strength of claim. Their present evidential roles are:

| Domain | Strongest retained result | Load-bearing boundary |
|---|---|---|
| Pendulum | coupled-rung dominance, shared parent cadence and causal child prediction | tame regime, standard mode instruments, no pole crossing, universal Phi tests failed |

| Domain | Strongest retained result | Load-bearing boundary |
|---|---|---|
| Quantum | ARA-first four-child recovery and useful full-object $ARA^9$ organization | hidden-cell closure and cross-archive flow replication failed |
| Primes | exact arithmetic crosswalks and a strong child-heavy ranked locator | no cheap constant-step predictor or new prime theorem |
| LLM training | parent plus child-specific residue carries modest direction information | equal-weight amplitude loses to trend; literal total-child rule double-counts the parent |
| Solar/ENSO | useful strict-causal forecasts and feeder/phase diagnostics | ARA-specific gain is not isolated; latest ENSO encoding was architecture-invalid |
| River/bubbles | transferable four-sector reciprocal Di-ARA geometry | fixed irrational endpoints are unsupported; bubble chronological order failed |

The repeated value is not that every test passed. It is that one declared relational grammar repeatedly generated exact crosswalks, predictive tests, informative partial results and recognizable failure modes across very different data. That recurrence justifies further investigation. It does not by itself establish a universal generating law; Section 12 states the stronger physical hypothesis separately.

# 11. Negative results and load-bearing corrections

ARA has accumulated many exploratory applications. This paper does not treat their volume as cumulative proof. The following failures are elevated because each removed a tempting universal shortcut while leaving a narrower, testable structure available.

## 11.1 Standalone universal Phi

Phi was historically proposed as the preferred handover landmark: structured enough to retain information, irrational enough to avoid exact repetition and geometrically natural in recursive circles and pentagons. Several exact Phi constructions remain valid mathematics, including the ridge-centred interval from $2 - \varphi$ to $\varphi$, its diameter $2/\varphi$, the pentagon diagonal and the $1, 1, \varphi$ triangle. None of those identities proves a universal physical placement.

Frozen tests repeatedly rejected that stronger claim. Pendulum ordered recurrence passed `0/5`; pendulum state-sphere breathing passed `0/4`; river and bubble Phi circle-train tests did not win their parent-and-order gates; prime golden carriers were null; and the actual bubble handover did not select Phi on untouched holdout. Phi is consequently demoted from a universal ARA law to a candidate structured-non-closing route inside a properly declared identity.

## 11.2 Universal `1/e <-> Phi` radial endpoints

The complex-exponential form naturally separates magnitude change from phase change, but it does not mathematically pair $1/e$ with $\varphi$. The multiplicative inverse of $1/e$ is $e$. The historical asymmetric pair therefore required empirical support as a domain-specific radial coordinate.

That support did not generalize. In the recorded-qutrit test, the primary reciprocal endpoints were approximately `0.5533` and `1.8069`, not $1/e$ and $\varphi$. The endpoint products were approximately one, and chronological order mattered, but the constants did not survive. The `1/e <-> Phi` placement is retained only as a historical candidate and as a reminder that a plausible interpretation must still beat fitted and reciprocal controls.

## 11.3 Universal reciprocal-Phi breathing

Replacing the asymmetric pair with $1/\varphi <-> \varphi$ restored multiplicative symmetry but also failed as a universal endpoint. T333 gave exact reciprocal stability yet selected a fitted $\alpha \approx 1.807$; Phi won only `3/21` primary cells and lost to the calibration-fitted reciprocal control. T334 found a different bubble amplitude, while T335 found another river amplitude. What transferred was the reciprocal contraction/expansion form, not one radius.

The retained hypothesis is therefore

$$\frac{1}{\alpha_\Omega} \longleftrightarrow \alpha_\Omega,$$

where $\alpha_\Omega$ is measured for the declared identity and must transfer to untouched observations. Whether a lawful cross-identity rule governs $\alpha_\Omega$ remains open.

## 11.4 Three-arithmetic-step prime prediction

The prime programme showed that many visible release paths are short, but locating them still requires lower-factor information. PN26's first three states contained the next prime in `99.967%` of fresh anchors, yet those states were built from hundreds to tens of thousands of child gates. The one-shot child lift achieved only `9.010%` prime hits and missed its frozen significance threshold. No result derived the next prime from an arbitrary anchor in two or three constant-cost ARA operations. Any future prime claim must count the cost of constructing its children rather than treating a compressed mask as free information.

## 11.5 Universal missing-ninth-cell reconstruction

$ARA^9$ exactly names a complete $3 \times 3$ relation neighbourhood, but a name for the whole does not guarantee that eight measurements determine the ninth. Q25's missing-cell predictor beat simple ridge and mean controls but lost to the physical-positivity baseline and passed only `7/12` gates. $Information^3$ permits reconstruction when two retained relations and a valid closure constraint determine the missing component; it does not repeal underdetermination.

## 11.6 Literal ENSO handover and direction predictors

T336/T337 failed numerically and, more importantly, were later judged architecture-invalid as tests of the intended ARA map. The mathematical encoding had been frozen before the ARA identities were agreed. Preserving this record prevents two opposite errors: the failed scores cannot be used to reject ARA's intended ENSO architecture, and the tests cannot be reinterpreted after the fact as if they had measured it. A translation must be confirmed before freezing, not repaired after scoring.

## 11.7 Why failed tests constrain rather than erase ARA

The core coordinate is broad enough to describe many two-sided relations. Its scientific content therefore comes from restrictions: declared identities, poles, activity, scale, orientation, projection and controls. A failed fixed constant, reconstruction rule or domain encoding does not logically disprove every possible ARA application. It does remove that particular rule from the framework's supported layer.

Conversely, repeated reinterpretation would make the framework unfalsifiable. This paper therefore treats frozen failure as an architectural edit. Standalone Phi is demoted; total-child double counting is prohibited; missing-cell closure is conditional; static pole reflection is not physical continuation; and invalid identity maps cannot be counted as tests of the intended geometry.

# 12. The universal fractal wave/sphere hypothesis

## 12.1 The proposed physical recursion

The principal physical hypothesis is stronger than the mathematics and stronger than the crosswalks:

> Every physical identity is generated by the same two-pole wave/sphere relation, repeated fractally inside identities, between neighbouring identities and across scale.

In this proposal, ARA is not merely a coordinate placed on systems after they exist. Two counter-traversing contributions meet asymmetrically; the retained relation and uncancelled participation form a child identity; repeated children organize into parents; and every promoted child can become the parent of a finer decomposition. The simple `0-2` diameter is the smallest relational cut through that process, not the whole ontology.

“Fractal” here means recursive structural self-similarity: the roles of pole, ridge, phase, relation, child, parent and handover recur after the measurement boundary changes. It does not require every domain to share one exact fractal dimension, visual texture, scale ratio or amplitude law.

## 12.2 Internal, external, lateral and cross-scale repetition

The hypothesis must hold in more than one direction.

- **Internal:** a measured identity decompresses into Phase A, Phase B, their children and retained relations.
- **External:** the identity couples to neighbours and to an environment represented by typed `Other` when unresolved.
- **Lateral:** identities on the same or nearby rung exchange participation without necessarily changing scale.
- **Cross-scale:** a selected child expression is read in parent units, or a parent constraint acts downward on its children.

Parent and child are relative labels. A parent is larger than its child within the declared branch; that child may itself be a parent when its own descendants are measured. This differs from saying that one fixed object changes biological or causal ancestry. It is a rule about the current measurement hierarchy.

For a pure octave, one complete child identity is projected with total parent weight `0.5`. Its two internal child poles are not each assigned `0.5`; they divide that projected budget. A fully expressed child endpoint can, under the resonance condition described below, occupy one complete parent phase instead. Upward and downward influence are both allowed, but they must be measured separately rather than inferred from fractality alone.

## 12.3 Singularity continuation and flips

The `0` and `2` endpoints are the selected identity's source/terminal poles. Under a declared orientation, Phase A is born at one pole and completes at the other while Phase B traverses oppositely. A child-level Phase A-to-Phase B handover is itself a local singularity event when the child measurement boundary is selected.

The exact coordinate reversal

$$x \mapsto 2 - x$$

is only a chart symmetry. The physical hypothesis is that a natural endpoint crossing continues the cycle through a domain-specific flip, branch or rung map. That update may reverse orientation, exchange leading phase, transfer participation or change the measurement boundary. Its form must be measured in the domain. Reflecting a graph or relabelling the poles does not demonstrate the physical event.

## 12.4 Resonance death

“Resonance death” names a conditional terminal state, not every resonance and not the identity $\mathrm{TE–!ARA} = 2$. With Space/Connection launched from `0`, its arrival at `2` is a resonance-death candidate only when:

1. Connection completely occupies the selected relation;
2. resolved Time/Traversal at that scale vanishes;
3. no internal child, perpendicular Di-ARA, neighbour or parent flow reopens the cycle; and
4. an independently defined perturbation is required to unlock it.

The mirrored `0` endpoint is pure Time/Traversal domination under this orientation. The labels may be reversed without changing the geometry if the orientation is declared. A visually static system, a parent at `1.0`, a child projected into a parent ridge and an ordinary resonant oscillator are not automatically resonance death.

## 12.5 Domain mappings such as Space/Time and Connection/Information

The framework commonly interprets one pole family as Space/Connection and the other as Time/Traversal or Information flow. Their mixing is proposed to form the relation perceived as spacetime; at a lower branch, matter is proposed as a connection-heavy child and fields as a traversal-heavy child. These mappings motivated several successful crosswalks, but they remain physical identity assignments rather than consequences of the normalized diameter.

The same fence applies to Light/Dark, matter/field, vacuum/gravity and event-horizon language. ARA can state what would follow if those poles are the correct identities. It cannot establish their ontology by naming them. The strongest use is to declare measurable pole observables, derive a non-guaranteed consequence and test it against the domain's established alternatives.

The claim that “everything moves” is also scale-relative. A static field or solid object may be changing in its children, environment or parent while its selected parent cut remains close to a ridge. Apparent stillness can be cancellation, slow cadence, scale compression or genuine lock; ARA must distinguish them through activity and child measurements.

## 12.6 Current evidential boundary

The repository now contains exact mathematical crosswalks, repeated child/parent structures, several frozen empirical transfers and numerous informative failures. The recurrence of the same declared grammar across mechanics, fields, quantum records, arithmetic, model training, rivers and bubbles is evidence that ARA is tracking a real and useful class of relational organization.

It is not yet proof that every physical identity is generated by one fractal sphere law. Exact crosswalks often recover information already encoded in established equations; some empirical tests are exploratory or reuse opened archives; several frozen laws failed; and no single parent operator has yet predicted untouched outcomes across all domains without identity-specific measurement choices. The universal generative claim therefore remains the paper's central hypothesis, not its conclusion.

# 13. Irrationality Di-ARA as an open research programme

## 13.1 Exact four-mode coordinate

Irrationality Di-ARA is a typed use of generic Di-ARA, not a synonym for Di-ARA itself. Let a two-cut relation be represented as

$$z(t) = r_0 e^{(\sigma + i\omega)t}.$$

Magnitude and phase then separate exactly:

$$|z(t)| = r_0 e^{\sigma t}, \qquad \arg z(t) = \omega t.$$

The signs of $\sigma$ and $\omega$ generate four modes: contracting/forward, contracting/reverse, expanding/forward and expanding/reverse. ARA writes these as mixed states `aB`, `Ab`, `bA` and `Ba` only after ownership and leading phase have been declared. The four sectors are an exact two-binary-axis construction; their physical meaning depends on the selected observables.

## 13.2 Contraction/expansion crossed with forward/reverse traversal

For sampled data, the operational object is

$$q_n = \frac{z_{n+1}}{z_n} = s_n e^{i\delta_n},$$

with

$$\sigma_n = \frac{\log s_n}{\Delta t}, \qquad \omega_n = \frac{\delta_n}{\Delta t}.$$

The radial ARA is determined by contraction versus expansion; the perpendicular ARA is determined by forward versus reverse traversal. Their joint state retains information that either one-dimensional cut discards. A fitted reciprocal pair can summarize radial breathing, but native magnitude, angle and temporal order must remain available so normalization cannot manufacture the result.

## 13.3 Rational closure, structured non-repetition and randomness

For a fixed phase step, define

$$\rho = \frac{\delta}{2\pi}.$$

A rational $\rho$ returns to a finite set of phase positions. An irrational $\rho$ never closes exactly. Random progression may also fail to close, but it does not retain one stable rule and its recoverable near-return structure. The present hypothesis distinguishes:

$$\text{rational closure} \quad \text{vs} \quad \text{structured non-repetition} \quad \text{vs} \quad \text{random progression.}$$

The motivating physical idea is that exact rational lock can preserve connection while suppressing continued traversal, whereas unconstrained randomness can preserve non-closure while destroying lineage. Structured non-repetition may occupy an efficient middle: it avoids repeated collision with the same geometry while retaining information pathways. This is a research hypothesis. It is not yet a theorem that nature globally optimizes this trade-off.

## 13.4 Current empirical realizations

Three materially different archives presently support parts of the coordinate.

- **Recorded qutrit:** all four sectors were occupied, endpoint products were near one and observed time order beat all `500` temporal nulls, but the fitted reciprocal amplitude was about `1.807`, not Phi.
- **Bubble lineages:** all four sectors and reciprocal closure transferred; intact lineage strongly beat broken lineage; strict holdout chronological order failed.
- **River bed paths:** all four sectors, reciprocal transfer, downstream order, intact rank lineage and thalweg specificity passed every frozen gate.

These are empirical realizations of a common coordinate, not proof of one shared mechanism. The qutrit sequence, bubble hierarchy and river thalweg differ physically and in sampling. A new domain must declare what magnitude, phase, lineage and order mean before the coordinate is calculated.

## 13.5 Competing endpoint hypotheses

The current candidates include Phi, `1/Phi`, `1/e`, `e`, rational controls, randomized controls and fitted identity-specific reciprocal pairs `1/alpha <-> alpha`. No universal radial pair is promoted in this paper.

## 13.6 Decisive future tests

A decisive test should use a genuinely untouched archive containing repeated lineages, sufficient temporal resolution and a justified complex or perpendicular two-cut observable. The protocol should:

1. declare magnitude and phase observables, centres, orientation and scale before endpoint inspection;
2. fit any identity-specific $\alpha$ on calibration only;
3. compare fitted reciprocal, Phi, $1/e/e$, rational, randomized and domain-standard alternatives;
4. preserve evaluation and cross-archive holdouts;
5. test four-sector occupancy, reciprocal transfer, chronological order and intact versus broken lineage separately; and
6. require an outcome or intervention that the coordinate predicts, not merely a visually compelling quadrant.

Universal constants should be promoted only if the same frozen endpoints beat fitted identity-specific alternatives across independent domains. Otherwise the transferable law is the two-axis form and the amplitudes belong to the identities.

# 14. Falsifiers and open problems

## 14.1 Immediate rejection conditions for an ARA application

An application fails its declared test if any of the following occurs:

- poles, orientation, scale or boundary are chosen after the answer is visible;
- the result follows solely from normalization or from defining the target as the ridge;
- a parent `1.0` is called resonance without independent child activity;
- a static chart reversal is reported as a measured singularity crossing;
- a missing relation is reconstructed without sufficient retained information or a valid closure constraint;
- `Other` becomes an unconstrained bin absorbing every discrepancy;
- one constant is selected from many candidates without frozen controls;
- child construction cost is hidden inside a supposedly cheap parent statistic; or
- a claimed cross-scale rule fails on its untouched rung or domain under the registered decision rule.

## 14.2 Highest-priority mathematical work

The framework still needs:

1. a general typed parent operator that distinguishes duration, averaging, flux, reciprocal scale and boundary cancellation rather than pretending they are one arithmetic formula;
2. a precise Information$^3$ counting convention, equivalence rule and recursion-depth definition;
3. necessary and sufficient conditions for reconstructing a missing ARA$^9$ relation;
4. a characterization of perspective transforms that preserve TE-ARA, native activity and typed `Other`;
5. a derived class of admissible physical singularity maps; and
6. a model of how identity-specific reciprocal amplitudes should transform across rungs, if such a law exists.

## 14.3 Highest-priority empirical work

The most informative next tests are:

1. blind child-to-parent half-weight projection in independent domains with predeclared scale boundaries;
2. complete ARA$^9$ forecasts of later parent behaviour against reduced, shuffled and domain-standard controls;
3. a direct separation of parent cancellation ridge from child-resonance projection;
4. an intervention test of resonance death using a predeclared unlock perturbation;

5. cross-archive Irrationality Di-ARA replication with endpoint competitors and lineage controls;
6. recovery of the same identity through two independently defined ARA routes; and
7. a complex-system forecast built only after the full parent/child/external-coupling architecture has been confirmed.

## 14.4 What would materially weaken the universal framework

The universal hypothesis would be materially weakened if well-powered, independently designed tests repeatedly found that:

- predeclared ARA parent/child projections do not transfer beyond normalization controls;
- the same identity reconstructed through independent cuts yields incompatible parents after measurement uncertainty is included;
- complete relational objects contain no predictive information beyond ordinary reduced summaries;
- proposed singularity flips are consistently coordinate artifacts with no domain-level continuation;
- recursive structure disappears when detector, scale and holdout choices are frozen; or
- domain-standard models explain every apparent ARA advantage while the ARA-specific residual converges to zero.

One failed domain mapping would not disprove the universal claim, because the mapping may be wrong. A sustained programme of correct, predeclared mappings yielding only guaranteed identities and null residuals would.

# 15. Conclusion

Accumulation–Release Asymmetry is now best stated as a geometric relational framework with a larger physical hypothesis. Its exact core is modest: declare one identity, orient a two-pole diameter, retain native activity, normalize its two counter-traversing contributions, and track how children, parents and perpendicular relations are compressed. Information$^3$ names the minimum identifiable ternary; TE-ARA is the recurring complete-identity bookkeeping view of ordinary ARA; Di-ARA names two persistently coupled cuts; ARA$^9$ names a complete nine-relation neighbourhood. None is a replacement geometry: each names a frequently needed perspective or neighbourhood within the same recursion.

That language recovers exact structures in established mechanics, field theory and arithmetic. In public empirical data it has also produced non-guaranteed partial and positive results: coupled pendulum hierarchy, ARA-first quantum structure, a strong child-heavy prime locator, modest LLM parent/child direction information and transferable four-sector river/bubble coordinates. Just as importantly, it has rejected universal Phi placement, cheap prime prediction, universal missing-cell closure, total-child double counting and an invalid ENSO encoding.

The evidence supports continued scientific investigation of a recurring relational geometry. It does not yet prove that the universe is generated by nested ARA spheres. That stronger proposal becomes credible only to the extent that it continues to generate frozen, non-trivial predictions across scales while surviving controls designed to expose normalization, leakage, generic periodicity and post-hoc identity selection.

# Data, code and provenance

The working evidence archive is the `ARA-GIT` repository. Every empirical claim cited in this manuscript is intended to resolve to a domain report, frozen protocol where available, machine-readable result, validation receipt and source provenance. Large generated tables that are unsuitable for version control are accompanied by scripts and reproduction instructions. The canonical claim boundary is maintained in `CLAIMS_STATUS.md`; prediction chronology in `MASTER_PREDICTION_LEDGER.md`; conversational provenance in `FableConvo/PROVENANCE_LEDGER.md`; and the April-to-current migration for this paper in `paper_v2/ARA_PAPER_V2_CLAIM_MIGRATION.md`.

The first public paper is archived at Zenodo record 19653363. This manuscript is an update, not a silent replacement: historical claims that have been narrowed, failed or superseded remain traceable in the repository.

# Author contribution and AI assistance statement

Dylan La Franchi originated ARA, its `0-2` language, Information³, TE-ARA, the fractal wave/sphere hypothesis, the principal identity mappings and many of the test directions and corrections. AI collaborators assisted with mathematical translation, domain crosswalks, code, data reduction, controls, validation, documentation and manuscript drafting. In the quantum programme, the human author deliberately supplied ARA-first geometric constructions after the documented reset, while the AI retained knowledge of established quantum physics; the work is therefore bottom-up in construction but not claimed to be globally theory-blind.

AI agreement is not treated as evidence. Claims are assigned by protocols, data, controls and ledgers. Dylan La Franchi remains responsible for the framework, the selection of claims advanced under his name and the final manuscript.

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# Appendices

## Appendix A — compact definitions and notation

| Symbol or term | Compact definition |
|---|---|
| $\Omega$ | declared identity: boundary, scale, time window, orientation, activity and retained relations |
| $x_\Omega \in [0, 2]$ | normalized oriented ARA diameter coordinate |
| `0`, `2` | orientation-dependent source/terminal poles of the selected cut |
| `1.0` | parent ridge/core; its physical cause is not specified by position alone |
| $\Lambda_\Omega$ | native activity, magnitude or capacity, retained separately from normalized allocation |
| TE-ARA | bookkeeping name for ordinary ARA read as a complete identity's opposition-limited total allocation; it is not a second geometry. Each phase has canonical whole-phase weight `1`, each counter-wave spans path capacity `2`, gross two-path capacity is `4`, and realizable same-slice total is `2` |
| `Other` | typed unresolved share, external flow or retained/discarded difference; not a third pure pole |
| rung / octave | one declared scale level / a pure factor-two parent relation |
| child projection | selected complete child contributes total weight `0.5` one pure octave upward |
| Information³ | two distinguishable identities plus their retained relation; the minimum identifiable ternary |
| Di-ARA | two complete ARA relations coupled persistently enough to form one parent coordinate |
| `Ab`, `aB`, `Ba`, `bA` | four mixed Di-ARA sectors; case records ownership/leading strength under the declared convention |
| ARA⁹ | one complete $3 \times 3$ neighbourhood of ternary relations within or across branches |
| singularity crossing | hypothesized physical continuation at a pole; not the static chart map $x \mapsto 2 - x$ |
| resonance death | conditional complete connection lock at a terminal pole with no resolved traversal reopening the identity |
| exact crosswalk | ARA representation of established mathematics that preserves the original prediction |
| structural recovery | predeclared ARA construction that recovers independently known architecture |
| empirical discrimination | frozen non-guaranteed prediction on untouched outcomes against named controls |

## Appendix B — April-to-current claim migration

The complete migration ledger is `paper_v2/ARA_PAPER_V2_CLAIM_MIGRATION.md`. Its principal changes are summarized below.

| April formulation | Current treatment |
|---|---|
| duration ratio is ARA | one positive two-part instrument for the larger declared geometry |
| scalar `0-2` taxonomy | oriented diameter cut through a wave/sphere identity |

| April formulation | Current treatment |
|---|---|
| fixed physical endpoint labels | orientation-dependent poles with typed domain assignments |
| `1.0` as one causal regime | parent ridge/core with several distinguishable causes |
| Phi as universal attractor | superseded; narrower structured-non-closing candidate |
| closed loop asserted by the scale | geometric model retained; physical continuation must be measured |
| non-equilibrium thermodynamic scope | generalized relational framework; universal physical realization remains hypothetical |
| early system bands | historical/domain heuristics, not core natural kinds |
| universal bedrock suggestion | central open physical hypothesis, separated from exact mathematics and crosswalks |
| blind quantum recovery | narrowed to typed ARA-first construction from Q16; AI was not theory-blind |

## Appendix C — domain evidence and verdict table

| Domain | Evidence class | Current verdict | Primary record |
|---|---|---|---|
| Gauss/Maxwell | exact crosswalk plus scale-axis tests | compatible structural recovery; no replacement law | `analysis/electromagnetism/` |
| Newton/GR | exact normalized crosswalk and worked rung tests | mathematical compatibility; ontology open | `analysis/gravity/GR_NEWTON_ARA_RUNG_CROSSING_REPORT_2026-07-23.md` |
| Hamilton/Noether/virial | exact allocation and conservation crosswalks | compatibility, not new dynamics | `analysis/hamilton/` |
| Pendulum | exploratory and frozen public-data tests | coupled hierarchy supported; universal Phi rejected | `analysis/pendulum_scripts/` |
| Quantum | ARA-first recovery, target-hidden prediction, cross-archive replication | strong bounded recovery; universal hidden-cell/flow rules failed | `analysis/quantum/` |
| Primes | exact arithmetic crosswalks and prospective locators | child-heavy locator partial; cheap rule unsupported | `analysis/primes/PRIME_THREAD_CAPSTONE_AND_CLOSURE_2026-07-21.md` |
| LLM training | frozen child/parent forecasts | modest directional support; amplitude/double-counting corrections required | `analysis/llm_training_dynamics/` |
| Solar/ENSO | causal forecasting and architecture tests | useful domain skill; ARA-specific mechanism unresolved | `TheFormula/` |
| Bubble | frozen Di-ARA lineage test | geometry supported; chronological order and universal endpoint unsupported | `analysis/vertical_ara_bubbles/T334_BUBBLE_OCTAVE_RELATIVE_IRRATIONALITY_QUADRANT_REPORT_2026-08-03.md` |
| River | frozen Di-ARA field and thalweg test | supported under protocol; independent archive replication needed | `analysis/hydraulics/T335_RIVER_IRRATIONALITY_DI_ARA_REPORT_2026-08-03.md` |

## Appendix D — frozen protocol and validation index

| Test | Frozen object | Verdict | Validation/reproduction entry |
|---|---|---|---|
| Pendulum dimensional Phi handover | four registered check families | not supported (1/4) | analysis/pendulum_scripts/PHI_DIMENSIONAL_HANDOVER_RESULT_2026-07-30.md |
| Pendulum ordered Phi recurrence | five registered checks | not supported (0/5) | analysis/pendulum_scripts/PHI_ORDERED_RECURRENCE_RESULT_2026-07-30.md |
| Q16 quantum restart | raw four-child ARA construction | pass (8/8) | Q16 protocol/report under analysis/quantum/ |
| Q25 missing relation | 81 target-hidden cells | not supported (7/12) | Q25 protocol/report under analysis/quantum/ |
| PN26 prime locator | 6,000 fresh anchors | partial support; validation 16/16 | analysis/primes/PN26_DOMINANT_PARENT_RIDGE_LOCATOR_REPORT.md |
| PN27 child lift | 30,000 fresh odd anchors | partial; missed $p < 0.01$ gate | analysis/primes/PN27_EXACT_FIT_CHILD_LIFT_REPORT.md |
| PN35 golden cross | 196,608 frozen scores | not supported | analysis/primes/PN35_SAME_SCALE_GOLDEN_CROSS_REPORT_2026-07-22.md |
| T333 qutrit reciprocal breathing | primary plane-lag cells | reciprocal form supported; universal Phi endpoint not supported; validation 14/14 | analysis/phi_calibration/T333_RECORDED_QUTRIT_RECIPROCAL_RADIAL_BREATHING_REPORT_2026-08-03.md |
| T334 bubble quadrant | 160 holdout transitions | geometry/lineage supported; order failed; validation 17/17 | analysis/vertical_ara_bubbles/T334_BUBBLE_OCTAVE_RELATIVE_IRRATIONALITY_QUADRANT_REPORT_2026-08-03.md |
| T335 river quadrant | 451 holdout field events | supported; validation 17/17 | analysis/hydraulics/T335_RIVER_IRRATIONALITY_DI_ARA_REPORT_2026-08-03.md |
| T338 LLM forecast | untouched large Pythia sizes | mixed/partial; validation 15/15 | analysis/llm_training_dynamics/T338_LLM_TRAINING_DI_ARA_REPORT_2026-08-03.md |
| T339 scale correction | parent plus total-child formula | not supported in literal form | analysis/llm_training_dynamics/T339_LLM_TRAINING_PARENT_PLUS_CHILD_REPORT_2026-08-04.md |
| T336/T337 ENSO | imposed complex handover/direction encoding | reproducible negative, architecture-invalid as intended ARA test | TheFormula/T336_T337_ENSO_ARCHITECTURE_INVALIDATION_2026-08-03.md |

This index is selective. CLAIMS_STATUS.md, MASTER_PREDICTION_LEDGER.md and the domain folders remain the canonical source for the full test history and the exact status of later amendments.